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Stochastics and Statistics

A study of data-driven distributionally robust optimization with incomplete joint data under finite support

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ABSTRACT

Missing data is a common issue for many practical data-driven stochastic programming problems. The state-of-the-art approaches first estimate the missing data values and then separately solve the corresponding stochastic programming. Accurate estimation of missing values is typically inaccessible as it requires enormous data and sophisticated statistical methods. Therefore, this paper proposes an integrated approach, a distributionally robust optimization (DRO) framework, that simultaneously tackles the missing data problem and data-driven stochastic optimization by hedging against the uncertainties of the missing values. This paper adds to the DRO literature by considering the practical scenario where the data can be incomplete and partially observable; it particularly focuses on data distributions with finite support. We construct several classes of ambiguity sets for our DRO model utilizing the incomplete data sets, maximum likelihood estimation method, and different metrics. We prove the statistical consistency and finite sample guarantees of the corresponding models and provide tractable reformulations of our model for different scenarios. We perform computational studies on the multi-item inventory control problem and portfolio optimization using synthetic and real-world data. We validate that our method outperforms the traditional estimate-then-optimized approaches.

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1. Introduction

This paper aims to address the issue of missing data (incomplete data) in data-driven stochastic programming. Stochastic programming is an important framework for optimization under uncertainty. It generally assumes that a probability distribution of the random variable $\boldsymbol{\xi}$ is available and seeks an optimal solution in terms of expected performance. In practice, the distribution of $\boldsymbol{\xi}$ is often unknown; consequently, data-driven approaches have been proposed to solve this problem. These data-driven methods only work well if well-conditioned historical data for $\boldsymbol{\xi}$ are available. However, if $\boldsymbol{\xi}$ is multidimensional, complete joint data of $\boldsymbol{\xi}$ are often hard to obtain in many real-world settings due to the following issues:

- Missing data in some dimensions for $\boldsymbol{\xi}$.
- Sharing of data is limited among dimensions representing different components.
- Different sizes of data in different dimensions.

Missing data is a common issue for practical operations research (OR) problems as data sets for many applications are incomplete. For example, a common challenge in portfolio optimization theory is that the historical data of the returns for assets have missing values (Radulescu, 2013; Taylor, 2006). The data in many other application domains are also incomplete, including large-scale transportation management systems (TMSs) used to monitor traffic conditions to improve traffic congestion. The mobility monitoring program of the Texas Transportation Institute (TTI) reports that after screening erroneous data, TMS data archives can be anywhere from 16% to 93% complete (Smith, Scherer, & Conklin, 2003).

The most popular method to solve the missing data problem is data imputation. In this approach, missing values are imputed before implementing optimization and performing other analyses on the complete data set. Many variants of data-imputation are proposed in the literature (Dempster, Laird, & Rubin, 1977; Little & Rubin, 2019; Stekhoven & Bühlmann, 2012; Troyanskaya et al., 2001; Wang, Li, Jiang, & Feng, 2006). In general, machine learning tools are intended to minimize prediction error and do not consider how the predictions will impact the downstream optimization problem Elmachtoub & Grigas (2022). Utilizing data-imputation methods for data-driven stochastic optimization facing incomplete data suffers from two major issues. First, estimate-then-optimize methods are known to give sub-optimal solutions in many recent studies (Delage & Ye, 2010; Esfahani & Kuhn, 2018; Gupta & Rusmevichientong, 2021; Liyanage & Shanthikumar, 2005). The paper (Liyanage

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& Shanthikumar, 2005) analyzes the newsvendor inventory control problem with an ambiguous demand. It shows the estimate-the-optimize approach leads to a suboptimal inventory policy. The authors present a better solution by integrating the estimation and the optimization tasks. Second, theoretical guarantees are hardly obtained; this is because the analysis of the missing data and the derivation of the stochastic optimization's optimal solutions are conducted separately.

We propose an integrated approach, a distributionally robust optimization (DRO) framework, that combines the estimation and the optimization steps for data-driven stochastic programming facing incomplete data. Compared with the existing missing data literature, our approach does not attempt to find a suitable estimation for the missing values. The goal of the presented method is to solve the corresponding data-driven stochastic programming. We quantify the uncertainties brought by the missing values and incorporate them into stochastic programming through the DRO framework.

This work also extends the DRO literature. DRO (Delage & Ye, 2010; Esfahani & Kuhn, 2018; Gao, Chen, & Kleywegt, 2017; Jiang & Guan, 2018; Zhao & Zhang, 2019) is a powerful modeling paradigm that combines the estimation and the optimization steps. This approach first constructs some ambiguity sets $\mathcal{P}$ based on the available data set; optimization techniques are then proposed to solve these models with respect to the worst-case distributions within the ambiguity sets. However, to the best of our knowledge, the existing DRO works have not considered any ambiguity sets based on the incomplete data sets. Researchers in Zhao & Zhang (2019) do consider a missing data problem encountered in incomplete trajectories data. But their main goal is to reconstruct the missing location-duration path choices, and their ambiguity set is still based on the complete historical data.

The main contributions of this paper are as follows: It proposes a DRO framework for data-driven optimization problems with incomplete (partially observable) data sets. We consider data distributions with finite support and assume only partially observed data are available, meaning that the components for each piece of data are randomly missing. We present several classes of ambiguity sets based on the incomplete data for our DRO framework and discuss their properties. We provide finite sample guarantees of our DRO model by utilizing the observed information matrix (Efron & Hinkley, 1978) into our analysis. We also prove the statistical consistency results using the properties of maximum likelihood estimation. Tractable reformulations of the models are presented. Finally, we conduct computational studies to evaluate the performances of the proposed approaches compared to data-imputation-based approaches based on both synthetic and real-world data. Below, we highlight the details of the contributions.

1. A new DRO framework based on incomplete data is proposed. The proposed DRO framework is fundamentally different from the popular data-imputation-based methods. It signifies an integrated model that solves the missing data problem and stochastic programming simultaneously instead of following the estimate-then-optimize procedure. By adopting a DRO framework, the presented models are robust towards the uncertainties of the missing values. Therefore, they greatly improve the out-of-sample performances in applications where the optimal solutions are sensitive to unknown parameters.
2. Our DRO framework extends the current studies on DRO by proposing ambiguity sets that are constructed directly based on the incomplete data set. We construct several types of ambiguity sets based on f-divergence, Wasserstein metric, L1 norm, and ellipsoids in the probability space. The centers of these ambiguity sets are chosen as optimal estimators of the maximum likelihood estimation (MLE) based on the incomplete data set. The first two kinds of ambiguity sets are inspired by two general metrics used in the DRO literature. The last one is inspired by the special structures in our model. We discuss that the ellipsoid ambiguity set is asymptotically optimal in the sense that it contains the complete data (true) distribution with the highest probability among all ambiguity sets having the same volume.
3. We obtain theoretical guarantees and tractable reformulations for the proposed models. We first derive the finite sample guarantees of our model by providing a probabilistic upper bound to their out-of-sample performances. Our analyses are based on the asymptotic normality and the observed information matrix (empirical fisher information). We then prove the statistical consistency guarantee, which means the solution of our model converges to the true optimal in probability when the number of observed data goes to infinity. Finally, we show that these reformulations can be efficiently solved if the cost functions of the original stochastic program are convex and the feasible regions are convex or mixed-integer linear sets.
4. We provide computational experiments for evaluation using both synthetic and real-world data. Two applications are studied: the multi-item inventory control problem and portfolio optimization. Considering the multi-item inventory control problem, we illustrate the reductions in the total costs of our models based on synthetic data. We benchmark our frameworks for portfolio optimization against the data imputation-based method using 57 pairs of training sets and test sets. These sets are obtained from real-world historical returns of exchange-traded funds (ETFs) and the US central bank (FED) rate of return from 2006 to 2016 (Boyd, 2019). We justify the conclusion of improvements by showing that proposed models consistently yield better out-of-sample performance.

The rest of the paper is organized as follows. We briefly review related works in Section 2. The background and notations are introduced in Section 3. We present the main results in Section 4. More specifically, in Section 4.1, we propose one ambiguity set, and we prove the corresponding statistical consistency and finite sample guarantees in Sections 4.2 and 4.3, respectively. We also discuss its reformulations in Section 4.4 and different classes of ambiguity sets in Section 4.5. Computational studies are summarized in Section 5. This paper concludes itself in Section 6. All the proofs and extensions to two-stage stochastic programming are provided in the Appendix.

2. Related works

This section briefly reviews related works about missing data and data-driven optimization under uncertainty.

Missing data. Missing or incomplete data has been studied widely, especially in machine learning and statistics (García-Laencina, Sancho-Gómez, & Figueiras-Vidal, 2010; Goodfellow, Bengio, & Courville, 2016; Rubin, 1976) literature. One of the most natural options to solve the missing data problem is to discard any data that include missing values. However, this approach may lead to biased results (Little & Rubin, 2019) or result in the loss of information.

For incorporating incomplete data into the decision-making process, the two most common and well-developed methods are data imputation and maximum likelihood-based approaches. The data imputation approaches recover the incomplete data set by estimating missing values based on the observed data. The criteria are often developed based on the decision trees (Stekhoven &

Bühlmann, 2012), support vector regression (Wang et al., 2006), neural networks (Nishanth & Ravi, 2016), etc. The maximum likelihood-based approaches ideally work for parametric models with missing but relatively complete data. These techniques often aim to find a distribution that maximizes the observed-data likelihood (Little & Rubin, 1983) or seeks EM algorithms (Dempster et al., 1977) because computational issues concerning non-convex optimization may arise (Ding & Song, 2016).

The statistical approaches developed above solely focus on estimating the missing values of the incomplete data sets. When these approaches are combined with stochastic optimization models, the performances are unclear. This study presents a new integrated model based on distributionally robust optimization. Instead of first estimating the unknown distribution from the incomplete data set, we propose integrated models to address the missing data issue and to find optimal decisions simultaneously. This allows our model to guarantee out-of-sample performances theoretically. We also demonstrate the effectiveness of this approach with computational experiments.

Data-driven optimization under uncertainty. Data-driven optimization has gained much attention recently because real-world data distributions are usually unknown. Sample average approximation (SAA) (Kleywegt, Shapiro, & Homem-de Mello, 2002; Shapiro & Homem-de Mello, 1998) approximates the unknown distribution through empirical distributions. If the data samples of true unknown distributions can be generated efficiently, the optimal solution of SAA converges to the true optimal with the probability 1. Stochastic approximation (SA) (Nemirovski, Juditsky, Lan, & Shapiro, 2009) is a similar approach compared to SAA and is also based on Monte Carlo sampling. It iteratively performs subgradient descent based on the observed samples to approach the true optimal solutions.

The min-max learning frameworks have gained popularity recently as they produce robust solutions that potentially improve out-of-sample performances. These frameworks appear in data-driven robust optimization (Bertsimas, Gupta, & Kallus, 2018a) and distributionally robust optimization (Delage & Ye, 2010; Erdoğan & Iyengar, 2006) problems. In these works, the ambiguity sets are constructed from the available data based on different ways, for example, moment-based set (Delage & Ye, 2010), confidence-region-based set (Bertsimas, Gupta, & Kallus, 2018b), metric-based set (Erdoğan & Iyengar, 2006; Jiang & Guan, 2016; Royset & Wets, 2017) and others (Chen, Sim, & Xu, 2019; Wang, Glynn, & Ye, 2016). These ideas are also extended and explored in two-stage stochastic decision-making schemes (Chen, Sim, & Xiong, 2020; Hanasusanto & Kuhn, 2018; Jiang & Guan, 2018; Zhao & Guan, 2018).

Thus far, all approaches discussed focus on analyzing the complete joint data of $\boldsymbol{\xi}$. When the data are incomplete (partially observable), the efficacy of these approaches is unclear. In this paper, we build our model based on the incomplete data set, providing valid solutions to the commonly encountered missing data problems.

In the following section, we introduce the background and notations. We present the main results and computational studies in Section 4 and Section 5.

3. Notations and background

Throughout this paper, we use the following notations and assumptions. A stochastic program can be formulated as

$$\min_{\mathbf{x}\in\mathcal{X}} \quad \mathbb{E}[Q(\mathbf{x},\boldsymbol{\xi})], \tag{1}$$

where $Q(\mathbf{x},\boldsymbol{\xi})$ represents a cost function with respect to decision $\mathbf{x}$, and $\mathcal{X}$ represents the feasible region of $\mathbf{x}$. In this paper, we study the problem whose random variable $\boldsymbol{\xi}$ has known finite support. Finite discrete support sets are common and studied in many operations research problems (Barbarosolu & Arda, 2004; Feng & Ryan, 2013; Rujeerapaiboon, Schindler, Kuhn, & Wiesemann, 2018; Zhang, Ahmed, & Lan, 2019). For example, in portfolio selection (Markowitz, 1959), decision-makers want to select a portfolio from a finite number of scenarios about possible returns to minimize some disutility function. Additionally, for problems with continuous random variables, methods like scenario construction (Shapiro, Dentcheva, & Ruszczyński, 2014) are often used to select some representative scenarios to simplify the problems. This also makes those random variables have known finite discrete support sets. Finally, we assume set $\mathcal{X}$ is not empty, and $Q(\mathbf{x},\boldsymbol{\xi})$ is bounded for $\mathbf{x}\in\mathcal{X}, \boldsymbol{\xi}\in\Xi$ to avoid discussing trivial cases.

We assume $\boldsymbol{\xi}$ is an I-dimensional vector $\boldsymbol{\xi}\in\Xi\subset\mathbb{R}^I$, and we refer to ξ_i as the ith *component*. If an observation of $\boldsymbol{\xi}$ is incomplete meaning that the values of some components are missing, it is very difficult to represent these missing values using the notation $\boldsymbol{\xi}$. To this goal, we define two new vectors $\mathbf{s}$ and $\mathbf{s}_{obs}$ to clearly index complete and incomplete data. We first explain the vector $\mathbf{s}$ representing the complete data of $\boldsymbol{\xi}$ as follows. Because the support of $\boldsymbol{\xi}$ is finite, we index each possible value of ξ_i with s_i which is a natural number, i.e. $s_i\in\{1,\ldots,J\}$ without loss of generality (J is a positive number). Correspondingly, this gives an I-dimensional index vector $\mathbf{s}\in\mathbb{N}^I$, which has one-to-one correspondence with $\boldsymbol{\xi}$. Here, $\mathbb{N}$ represents the set of natural numbers. Therefore, each $\mathbf{s}$ represents one *scenario* of $\boldsymbol{\xi}$. The set containing all scenarios is denoted as $\mathcal{S}$, and its cardinality is $|\mathcal{S}|$. We define another vector $\mathbf{s}_{obs}\in\{0,\mathbb{N}\}^I$ representing the incomplete data. It follows the same definition as $\mathbf{s}$ except that its components can take the value of zero. If its ith component equals zero, i.e., $s_{obs,i}=0$, it indicates that the value of the ith component is missing. We want to emphasize that our model treats the zeros as categorical data. The above procedure does not mean we replace all missing values with 0.

In the rest of the paper, we use $\mathbf{s}$ and $\boldsymbol{\xi}$ interchangeably. We use the following common signs for different types of convergence of random variables. (1) Converge in probability: $\xrightarrow{p}$. (2) Converge almost surely: $\xrightarrow{a.s.}$. (3) Converge in distribution: $\xrightarrow{d}$.

Terminologies in data-driven stochastic programming. We briefly review some commonly used terminologies and notations in data-driven stochastic programming.

We use $\mathbf{P}=\{\mathrm{P}(\mathbf{s}),\forall\mathbf{s}\in\mathcal{S}\}$ to denote a joint distribution of $\mathbf{s}$, and $\mathbf{P}^*=\{\mathrm{P}^*(\mathbf{s}),\forall\mathbf{s}\in\mathcal{S}\}$ represents the true unknown joint distribution of $\mathbf{s}$. We call $\mathbf{P}^*$ the *true distribution*. Then the *true optimal cost*, O^*, is defined as

$$O^*=\min_{\mathbf{x}\in\mathcal{X}} \quad \mathbb{E}_{\mathbf{P}^*}[Q(\mathbf{x},\mathbf{s})],$$

where the expectation is taken with respect to the true distribution $\mathbf{P}^*$. Any corresponding optimal solution $\mathbf{x}^*$ is called the *true optimal solution*.

Suppose a data set of $\mathbf{s}$ contains N i.i.d. data points in total. A *data-driven solution* for (1) is defined as a feasible solution $\hat{\mathbf{x}}_N\in\mathcal{X}$ based on this data set. The *out-of-sample performance* of $\hat{\mathbf{x}}_N$ is defined as

$$\mathbb{E}_{\mathbf{P}^*}[Q(\hat{\mathbf{x}}_N,\mathbf{s})]. \tag{2}$$

However, the out-of-sample performance cannot be directly evaluated because $\mathbf{P}^*$ is unknown. Therefore, we seek an upper bound $\hat{O}_N$ to obtain the performance guarantees of the type

$$\mathrm{P}\big(\mathbb{E}_{\mathbf{P}^*}[Q(\hat{\mathbf{x}}_N,\mathbf{s})]\leqslant\hat{O}_N\big)\geqslant 1-\alpha, \tag{3}$$

where $\alpha\in(0,1)$. We use P to denote a product probability measure based on the training data set because all the randomness comes from the observed data. We refer to $1-\alpha$ as the *reliability*, which measures the probability that the out-of-sample performance is bounded by $\hat{O}_N$. We will explicitly define $\hat{O}_N$ later. The corresponding Eq. (3) is directly referred to as the *finite sample guarantee* throughout the rest of the paper.

3.1. Distributionally robust optimization framework with missing data

Our DRO model begins by defining an ambiguity set $\mathcal{P}$ for the unknown joint distribution based on the observed incomplete data set; $\mathcal{P}$ contains different possible joint distributions $\mathbf{P}$ and includes the true distribution $\mathbf{P}^*$ with a prescribed probability, $\mathrm{P}(\mathbf{P}^* \in \mathcal{P}) \geqslant 1-\alpha$, for $0 < \alpha < 1$. The parameter α can be used to reflect the conservatism of the model. After determining $\mathcal{P}$, a distributional robust approach finds the best solution, assuming the worst-case distribution within $\mathcal{P}$ as shown in (4),

$$\min_{\mathbf{x}\in\mathcal{X}} \max_{\mathbf{P}\in\mathcal{P}} \sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s}) Q(\mathbf{x}, \mathbf{s}). \tag{4}$$

The key difference of the DRO framework studied in this paper is that $\mathcal{P}$ uses incomplete data sets and addresses the uncertainties associated with the incomplete data set. We point out the benefits of our model comparing to the other methods of solving missing data problems as follows.

Benefits of model (4). Compared to the traditional approaches of solving missing data problems, Model (4) is unique in that it integrates the missing data directly into the final decision-making process. This is achieved by combining the information of the incomplete data set and the derivation of $\mathbf{x} \in \mathcal{X}$ into one optimization problem. Classical methods use separate analysis (Enders & Bandalos, 2001; Sterne et al., 2009), which first estimates $\mathbf{P}$ and then derives the optimal $\mathbf{x}$. It is recognized in literature (Gupta & Rusmevichientong, 2021; Liyanage & Shanthikumar, 2005) that separate analyses ignoring the potential estimation errors can lead to sub-optimal solutions and/or heuristic methods with no performance guarantees.

In the following, we propose a method to construct the ambiguity set based on the incomplete data set; the obtained optimal decision from the incomplete data allows the theoretical guarantees and improves the performance in practical problems.

4. Main model

In this section, we first introduce and discuss our model based on one specific ambiguity set. Then, we introduce the concepts of asymptotic normality and the observed information matrix to obtain the theoretical guarantees, which contain two main points. First, we prove the consistency result in Section 4.2. The consistency follows the definition in Van der Vaart (2000) meaning that the solution of our model converges in probability to the *true optimal solution* as the observed data size N goes to infinity. Second, we investigate the finite sample guarantee (3) in Section 4.3, where it shows the objective value of our model serves as a probabilistic upper bound of the out-of-sample performance. Finally, we discuss several other kinds of ambiguity sets with tractable reformulations in Section 4.4.

Assumption and notations. We recall the setting of the incomplete data and make one technical assumption. The incomplete data, or more intuitively the partially observed data, means that the values of the components are missing at random, and the number of observed dimensions for each data point ranges randomly from 1 to I. Recall that vector $\mathbf{s}_{obs} \in (\{0\}, \mathbb{N})^I$ can be used to represent the partially observed data, where its ith component $s_{obs,i} = 0$ indicates that the value of the ith component is missing. Correspondingly, we further define an indicator vector $\boldsymbol{\phi}_{obs} \in \{0,1\}^I$ to indicate the observed dimensions of $\mathbf{s}_{obs}$ ($\phi_{obs,i} = 1$ represents the ith dimension is observed; 0 otherwise). Finally, for a fixed $\mathbf{s}_{obs}$ we define $\mathcal{S}(\mathbf{s}_{obs})$ as a set that

$$\mathcal{S}(\mathbf{s}_{obs}) = \{\mathbf{s} \in \mathcal{S} | \mathbf{s} \odot \boldsymbol{\phi}_{obs} = \mathbf{s}_{obs}\}, \tag{5}$$

where $\odot$ represents the component-wise multiplication. Intuitively, set $\mathcal{S}(\mathbf{s}_{obs})$ contains all the complete data $\mathbf{s}$ that match the observed components in $\mathbf{s}_{obs}$. The nth partially observed data is denoted as $\hat{\mathbf{s}}_{n,obs}$ ($n = 1, \ldots, N$), which is the nth realization of $\mathbf{s}_{obs}$.

We make a common assumption on the missing data mechanism called missing at random (MAR) (Little & Rubin, 2019). This assumption means that the missing probability of a missed value is independent of this value itself. Mathematically, this means the probability of observing one $\mathbf{s}_{obs}$ given $\mathbf{s}$ is fixed for all $\mathbf{s} \in \mathcal{S}(\mathbf{s}_{obs})$ as shown below,

$$\mathrm{P}(\mathbf{s}_{obs}|\mathbf{s}) = \mathrm{P}(\mathbf{s}_{obs}|\mathbf{s}'), \quad \mathbf{s}, \mathbf{s}' \in \mathcal{S}(\mathbf{s}_{obs}). \tag{6}$$

If the missing probability of a component depends on the values of itself, the missing data mechanism itself has to be known or modeled explicitly to recover the missing values.

4.1. Ambiguity set $\mathcal{P}'$ based on partially observed data

We propose an ambiguity set $\mathcal{P}'$ centering around a nominal distribution ($\hat{\mathbf{P}}$) based on some metric $\|\dot{\|}$ as shown in (7).

$$\mathcal{P}' = \{\mathbf{P} \text{ is a distribution function } : \|\mathbf{P} - \hat{\mathbf{P}}\| \leqslant \tau\}, \tau > 0. \tag{7}$$

Before explaining the details about (7), we want to point out that this nominal distribution is chosen as one optimal estimator of the maximum likelihood estimation (MLE) based on the incomplete data set. The ambiguity set $\mathcal{P}'$ has a user tunable parameter τ called *distance tolerance* to control the robustness. We will further prove the relationship between τ and out-of-sample performance. This ambiguity set enjoys several good properties. 1) The nominal distribution, or the center of the ambiguity set equivalently, represents a joint distribution that has the largest likelihood to generate the observed incomplete data set; 2) Ambiguity set $\mathcal{P}'$ is consistent with the popular metric-based/likelihood-based ambiguity sets (Erdoğan & Iyengar, 2006; Esfahani & Kuhn, 2018; Hu & Hong, 2013; Pflug & Wozabal, 2007). Because in these works, researchers define the center of the ambiguity set as the empirical distribution, which can be viewed as one optimal solution of (nonparametric) MLE (Owen, 2001); 3) We will show that $\mathcal{P}'$ allows us to obtain the finite sample guarantee of Model (4) directly through the distance tolerance and ensure the obtained solution to converge to the true optimal as data size increases even though the data are partially observed.

Formulation of $\mathcal{P}'$ with L1 norm. In what follows, we define the ambiguity set mathematically in (8) by using the L1 norm and will focus on this ambiguity set in the next two subsections. Other types of ambiguity sets will be further investigated in Section 4.5. The L1 norm is special in this problem because the random variables with known discrete support often represent categorical variables instead of ordinal ones, like the portfolio selection and data of transportation routes introduced before. The L1 norm is a meaningful norm here when the data is nominal. More details about this point are listed in Appendix A for interested readers.

$$\mathcal{P}' = \left\{ \mathbf{P} = \{\mathrm{P}(\mathbf{s}), \forall \mathbf{s} \in \mathcal{S}\} : \begin{array}{l} \sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s}) = 1, \\ \mathrm{P}(\mathbf{s}) \geqslant 0, \quad \forall \mathbf{s} \in \mathcal{S}, \\ \sum_{\mathbf{s}\in\mathcal{S}} |\mathrm{P}(\mathbf{s}) - \hat{\mathrm{P}}(\mathbf{s})| \leqslant \tau. \end{array} \right\} \tag{8}$$

Set $\mathcal{P}'$ follows the general definition in (7). It contains all the joint distributions that are close to a nominal distribution $\hat{\mathbf{P}}$, and a *distance tolerance* τ controls the conservatism. Intuitively, we call the nominal distribution $\hat{\mathbf{P}}$ as the *center* of the ambiguity set.

We obtain the center $\hat{\mathbf{P}}$ of the ambiguity set through MLE. In Proposition 1, we show that this procedure is equivalent to solving a convex optimization.

Proposition 1. *The nominal distribution $\hat{\mathbf{P}}$ is one optimal solution of the following optimization*

$$\begin{aligned} \max_{\mathbf{P}} \quad & \sum_{n=1}^{N} \ln \left[\sum_{\mathbf{s}\in\mathcal{S}(\hat{\mathbf{s}}_{n,obs})} P(\mathbf{s}) \right] \\ s.t. \quad & P(\mathbf{s}) \geqslant 0, \forall \mathbf{s}, \\ & \sum_{\mathbf{s}\in\mathcal{S}} P(\mathbf{s}) = 1, \end{aligned} \tag{9}$$

where we use $\mathcal{S}(\hat{\mathbf{s}}_{n,obs})$ to denote all the $\mathbf{s} \in \mathcal{S}$ that match the observed part of the nth data $\hat{\mathbf{s}}_{n,obs}$ as defined in (5).

Proof. Please refer to Appendix B for detailed proofs. □

Optimization (9) is a maximization of a concave function under linear constraints, which can be solved through convex optimization techniques like projected gradient descent (Boyd & Vandenberghe, 2004) or simply via general convex solvers like CVXOPT. We also discuss a simple example below to give some intuitions on the nominal distribution. We assume the support $\mathcal{S} = \{(1,1),(1,2),(2,2)\}$. If we observed two incomplete data $\{(1,0),(0,2)\}$ (recall 0 here denotes unobserved values because of the relabeling introduced at the beginning), the nominal distribution is set to $(1/4, 1/2, 1/4)$. This represents that the probability for (1,1) is 1/4, and the probability for (1,2) is 1/2. The probability of scenario (1,2) is the highest because it matches both incomplete observations. If the observed data are $\{(1,0),(2,0)\}$, the corresponding nominal distribution is $(1/4, 1/4, 1/2)$. Intuitively, because there is only one scenario whose first dimension equals 2, the incomplete data (2,0) can be viewed as a complete observation.

4.2. Consistency

In this section, we validate that the optimal solution of Optimization (4) under the ambiguity set (8) converges to the true optimal solution in probability as the data size increases to infinity. Intuitively, this result implies that the obtained optimal decision of our model is as good as a decision made under the complete information when the available data size is large enough.

Mathematically, we aim to prove that given the distance tolerance τ converging to zero, and the number of data samples, N, goes to infinity, the solution to the proposed DRO model converges in probability to the true optimal solution. We establish this desired result in Theorem 1. To prove Theorem 1, we rely on first proving Lemmas 1 and 2.

Intuitively, Lemma 1 states that a unique maximum exists for (9) when the data size $N \to \infty$, and the corresponding solution of this unique maximum is the true joint distribution $\mathbf{P}^*$.

Lemma 1. *Define $F(\mathbf{s}_{obs}|\mathbf{P}) = \sum_{\mathbf{s}\in\mathcal{S}(\mathbf{s}_{obs})} P(\mathbf{s})$. Let*

$$L(\mathbf{P}) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \ln F\left(\hat{\mathbf{s}}_{n,obs}|\mathbf{P}\right) = \mathbb{E}[\ln F(\mathbf{s}_{obs}|\mathbf{P})],$$

where the expectation is taken over the probability mass function of $\mathbf{s}_{obs}$. Then, $L(\mathbf{P})$ attains its maximum uniquely at $\mathbf{P}^*$, where $\mathbf{P}^*$ represents the true joint distribution.

Proof. Please refer to Appendix C for detailed proofs. □

Based on Lemmas 1 and 2 further proves that the center $\hat{\mathbf{P}}$ of the ambiguity set $\mathcal{P}'$ converges in probability to the true joint distribution $\mathbf{P}^*$ as the data size increases to infinity. The main idea behind Lemma 2 is to prove the uniform law of large numbers for the objective function of (9), which is achieved by constructing a dominating function of it.

Lemma 2. *When the number of samples, N, goes to infinity, the solution $\hat{\mathbf{P}}$ of (9) converges in probability to $\mathbf{P}^*$.*

Proof. Please refer to Appendix D for detailed proofs. □

Finally, we establish the consistency results by using Lemmas 1 and 2. We use $\tau(N)$ to imply that the distance tolerance τ is a function of the sample size N.

Theorem 1 (Consistency)**.** *Let $\tau(N)$ satisfy*

$$\lim_{N\to\infty} \tau(N) = 0.$$

Assume $\hat{\mathbf{x}}_N$ and $\hat{O}_N$ are one optimal solution and the corresponding objective value of the proposed model, i.e.

$$\min_{\mathbf{x}\in\mathcal{X}} \max_{\mathbf{P}\in\mathcal{P}'} \mathbb{E}_{\mathbf{P}}[Q(\mathbf{x},\mathbf{s})],$$

under the incomplete data set of size N. Let $\mathbf{x}^*$ and O^* be one optimal solution and the objective value of

$$\min_{\mathbf{x}\in\mathcal{X}} \mathbb{E}_{\mathbf{P}^*}[Q(\mathbf{x},\mathbf{s})],$$

where the expectation is taken with respect to the true unknown joint distribution. Then $\hat{O}_N \xrightarrow{p} O^*$. Furthermore, if the maximizer $\mathbf{x}^*$ is unique, then $\hat{\mathbf{x}}_N \xrightarrow{p} \mathbf{x}^*$.

Proof. Please refer to Appendix E for detailed proofs. □

In Theorem 1, if there are multiple solutions, the optimal objective value of our model still converges to the true optimal. This implies that we obtain some solution whose performance is as good as the true optimal.

4.3. Finite sample guarantee

In this section, we establish the finite sample guarantee that is defined in (3). We show that the optimal value of the objective function (4) serves as a probabilistic upper bound of the out-of-sample performance under the finite number of incomplete data. The main steps in our proofs are as follows. We first prove that the deviation between the true distribution and the nominal distribution can be approximated by a normal distribution asymptotically. This normal distribution has a mean zero and a variance-covariance matrix that depends only on the true distribution function and the missing data mechanism. Because we assume the true distribution function and the detailed missing probabilities of components are unknown, this variance-covariance matrix cannot be directly obtained. We propose to obtain it via observed information matrix (Dong & Yin, 2018; Efron & Hinkley, 1978), which can be viewed as a data-driven version of this variance-covariance matrix. Then, we obtain the finite sample guarantee based on the obtained asymptotic normal distribution.

Notations. We assume set

$$\mathcal{S}^+ = \{\mathbf{s} \in \mathcal{S} | \hat{\mathrm{P}}(\mathbf{s}) > 0\} = \{\mathbf{s}^{\{1\}}, \ldots, \mathbf{s}^{\{b\}}\}$$

contains b different $\mathbf{s}$. Their corresponding $\hat{\mathrm{P}}(\mathbf{s})$ are represented with a vector $\mathbf{p} = [p_1, \ldots, p_b]$; we also define $\{a_1, \ldots, a_b\}$ to be the corresponding number of observations for these b different $\mathbf{s}$ in the data set. Suppose we also observe q different incomplete data (the number of observed dimensions is less than I) $\mathbf{s}_{obs}^{\{j\}}$, $j = 1, \ldots, q$, and each $\mathbf{s}_{obs}^{\{j\}}$ appears b_j times in the data set. We define a vector $\boldsymbol{\delta}_j \in \{0,1\}^b$, $j = 1, \ldots, q$, where $\delta_{ji} = 1$ if the observed dimensions of $\mathbf{s}_{obs}^{\{j\}}$ match that of $\mathbf{s}^{\{i\}}$, i.e.

$$\mathbf{s}^{\{i\}} \odot \boldsymbol{\phi}_{obs}^{\{j\}} = \mathbf{s}_{obs}^{\{j\}}.$$

Recall that $\boldsymbol{\phi}_{obs} \in \{0,1\}^I$ was defined as the indicator vector of observed dimensions.

Proposition 2 states that for a fixed distance tolerance τ, the deviation between the true joint distribution $\mathbf{P}^*$ and the center $\hat{\mathbf{P}}$ of the ambiguity set is less than a distance tolerance τ with probability at least α, where α is a function of τ and the observed data. The main idea behind Proposition 2 is to show that the difference of $\mathbf{P}^*$ and $\hat{\mathbf{P}}$ can be well approximated by a random variable with distribution $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma^*})$. More details are provided in the appendix.

Proposition 2. *Suppose the true joint distribution is $\mathbf{P}^*$. The nominal joint distribution $\hat{\mathbf{P}}$ satisfies $\sum_{\mathbf{s}\in\mathcal{S}} |P^*(\mathbf{s}) - \hat{P}(\mathbf{s})| \leqslant \tau$, with probability at least $1-\alpha$, where*

$$\alpha = 1 - \int_{[-\frac{\tau}{b},\frac{\tau}{b}]^b} \frac{1}{\sqrt{(2\pi)^b|\mathbf{\Sigma}^*|}} \exp\left(-\frac{1}{2}\mathbf{x}^T\mathbf{\Sigma}^{*-1}\mathbf{x}\right)d\mathbf{x},$$

and $\mathbf{\Sigma^*} \in \mathbb{R}^{b\times b}$ is the asymptotic variance-covariance matrix of $\hat{\mathbf{P}} - \mathbf{P}^*$. In practice, we can derive $\mathbf{\Sigma^*}$ empirically, which is denoted as $\mathbf{\Sigma}$ below.

$$\mathbf{\Sigma} = \frac{1}{N}\begin{bmatrix} \mathbf{\Sigma}_{b-1} & \mathbf{v}_b \\ \mathbf{v}_b^T & var(b) \end{bmatrix}$$

with $\mathbf{\Sigma}_{b-1} \in \mathbb{R}^{b-1\times b-1}$, $\mathbf{v}_b \in \mathbb{R}^{b-1}$. For matrix $\mathbf{\Sigma}$, we define $var(b) = \mathbf{1}^T\mathbf{\Sigma}_{b-1}\mathbf{1}$ and $\mathbf{v}_b = \mathbf{\Sigma}_{b-1}\mathbf{1}$, where $\mathbf{1}$ denotes one $(b-1)$-dimensional column vector whose all elements equal to 1. In addition,

$$\mathbf{\Sigma}_{b-1}^{-1} = diag\left(\frac{a_1}{p_1^2},\ldots,\frac{a_{b-1}}{p_{b-1}^2}\right) + \frac{a_{b-1}}{p_{b-1}^2}\begin{bmatrix} 1 & \ldots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \ldots & 1 \end{bmatrix} + \begin{bmatrix} \psi_{11} & \ldots & \psi_{1,b-1} \\ \vdots & \ddots & \vdots \\ \psi_{b-1,1} & \ldots & \psi_{b-1,b-1} \end{bmatrix},$$

where

$$\psi_{ik} = \sum_{j=1}^{q} \frac{b_j(\delta_{ji}-\delta_{jb})(\delta_{jk}-\delta_{jb})}{(\boldsymbol{\delta}_j^T\mathbf{p})^2}, 1 \leqslant i,k \leqslant b-1.$$

Proof. Please refer to Appendix F for detailed proofs. □

Following the results of Proposition 2, we directly establish the finite sample guarantee, which is defined in (3), in Theorem 2 below.

Theorem 2 (Finite sample guarantee)**.** *Suppose $\mathbf{P}^*$ represents the true joint distribution, and $\hat{\mathbf{x}}_N$ and $\hat{O}_N$ are one optimal solution and the corresponding objective value of Model (4) with ambiguity set $\mathcal{P}'$ under N data. Then, we have*

$$P\left[\mathbb{E}_{\mathbf{P}^*}[Q(\hat{\mathbf{x}}_N, \mathbf{s})] \leqslant \hat{O}_N\right] \geqslant 1-\alpha,$$

where α follows the definition in Proposition 2.

We used the asymptotic approximation (similar to Jiang & Guan, 2018) in the proofs of the above Theorem. This provides theoretical justification for using the proposed model. In practice, it is advised to select the parameter τ according to the concrete problems and the available datasets (for example, via cross-validation) instead of using the conservative a priori bound Esfahani & Kuhn (2018). So far, we have shown that the proposed ambiguity set allows our DRO model with favorable asymptotic and finite sample guarantees when only incomplete data sets are available. In the next subsection, we show that this DRO model is also mathematically tractable for a variety of cases.

Table 1
Some f-divergence examples.

f-divergence	$D_f(\mathbf{P}\|\|\hat{\mathbf{P}})$
Kullback–Leibler	$\sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s})\log(\frac{\mathrm{P}(\mathbf{s})}{\hat{\mathrm{P}}(\mathbf{s})})$
Burg entropy	$\sum_{\mathbf{s}\in\mathcal{S}} \hat{\mathrm{P}}(\mathbf{s})\log(\frac{\hat{\mathrm{P}}(\mathbf{s})}{\mathrm{P}(\mathbf{s})})$
J-divergence	$\sum_{\mathbf{s}\in\mathcal{S}}[\mathrm{P}(\mathbf{s}) - \hat{\mathrm{P}}(\mathbf{s})]\log(\frac{\mathrm{P}(\mathbf{s})}{\hat{\mathrm{P}}(\mathbf{s})})$
χ^2-distance	$\sum_{\mathbf{s}\in\mathcal{S}} \frac{(\mathrm{P}(\mathbf{s})-\hat{\mathrm{P}}(\mathbf{s}))^2}{\mathrm{P}(\mathbf{s})}$
Hellinger distance	$\sum_{\mathbf{s}\in\mathcal{S}}(\sqrt{\mathrm{P}(\mathbf{s})} - \sqrt{\hat{\mathrm{P}}(\mathbf{s})})^2$
Total variation	$\sum_{\mathbf{s}\in\mathcal{S}} \|\mathrm{P}(\mathbf{s}) - \hat{\mathrm{P}}(\mathbf{s})\|$

4.4. Worst-case reformulation

In a classical stochastic program, the cost function $Q(\mathbf{x}, \mathbf{s})$ is linear with respect to $\mathbf{x}$, and the feasible region for $\mathbf{x}$ $(\mathcal{X})$ is a polytope. Then, solving our model is equivalent to solving a linear program. In general, if $Q(\mathbf{x}, \mathbf{s})$ is convex with respect to $\mathbf{x}$, and $\mathcal{X}$ is a convex set or mixed-integer linear set (Conforti, Cornuéjols, & Zambelli, 2014), our model is mathematically tractable. We conclude the results in Proposition 3.

Proposition 3. *Optimization*

$$\min_{\mathbf{x}\in\mathcal{X}} \quad \max_{\{P(\mathbf{s}),\forall\mathbf{s}\in\mathcal{S}\}\in\mathcal{P}'} \sum_{\mathbf{s}\in\mathcal{S}} P(\mathbf{s})Q(\mathbf{x},\mathbf{s})$$

is equivalent to

$$\begin{aligned} \min_{\mathcal{B}} \quad & \gamma + e\tau + \sum_{\mathbf{s}\in\mathcal{S}}(w_{\mathbf{s}} - l_{\mathbf{s}})\hat{P}(\mathbf{s}) \\ s.t. \quad & Q(\mathbf{x},\mathbf{s}) \leqslant \gamma + w_{\mathbf{s}} - l_{\mathbf{s}}, \forall\mathbf{s}\in\mathcal{S}, \\ & w_{\mathbf{s}} + l_{\mathbf{s}} - e = 0, \forall\mathbf{s}\in\mathcal{S}, \\ & \mathbf{x}\in\mathcal{X}, \end{aligned} \tag{10}$$

where $\mathcal{B} = \{\mathbf{x}, \gamma, w_{\mathbf{s}} \geqslant 0, l_{\mathbf{s}} \geqslant 0, e \geqslant 0\}$.

Proof. Please refer to Appendix G for detailed proofs. □

Finally, the above results can be extended to two-stage stochastic programming cases, where related results are summarized in Appendix H.

4.5. Other ambiguity sets

Other metrics can be used in (7) to construct the ambiguity set. These ambiguity sets enjoy the same asymptotic result (consistency) and similar finite-sample guarantees. This is because the center of the ambiguity set is fixed to be $\hat{\mathbf{P}}$ in (7), which implies the difference between the center and the true joint distribution can still be well approximated by a normal random variable asymptotically. We discuss several kinds of ambiguity sets in this subsection and their reformulations.

1. **f -divergence-based ambiguity set.** In this case, we define the distance between $\mathbf{P}$ and $\hat{\mathbf{P}}$, i.e., $\|\mathbf{P} - \hat{\mathbf{P}}\|$, to be an f-divergence (Lin, 1991), which is a function $D_f(\mathbf{P}||\hat{\mathbf{P}})$ that measures the difference between two probability distributions. We list some f-divergence examples below in Table 1. We want to point out that the L1 norm discussed before can also be viewed as a special case of f-divergence (total variation).

$$\mathcal{P}' = \left\{ \mathbf{P} = \{\mathrm{P}(\mathbf{s}), \forall\mathbf{s}\in\mathcal{S}\} : \begin{array}{l} \sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s}) = 1, \\ \mathrm{P}(\mathbf{s}) \geqslant 0, \quad \forall\mathbf{s}\in\mathcal{S}, \\ D_f(\mathbf{P}||\hat{\mathbf{P}}) \leqslant \tau. \end{array} \right\} \tag{11}$$

The DRO with f-divergence-based ambiguity sets is mostly tractable, which are presented in Ben-Tal, Den Hertog,

De Waegenaere, Melenberg, & Rennen (2013). We refer readers to it for tractable reformulations by replacing the center of the uncertainty set discussed in their paper with the nominal distribution defined in our paper. Finally, we can obtain a similar finite-sample guarantee based on the results in Proposition 2. We conclude it as follows.

Corollary 1 (Finite sample guarantee)**.** *Suppose* $\mathbf{P}^*$ *represents the true joint distribution, and* $\hat{\mathbf{x}}_N$ *and* $\hat{O}_N$ *are one optimal solution and the corresponding objective value of Model (4) with ambiguity set based on f-divergence* $D_f(\mathbf{P}||\hat{\mathbf{P}})$ *under N data. Then, we have*

$$P\Big[\mathbb{E}_{\mathbf{P}^*}[Q(\hat{\mathbf{x}}_N, \mathbf{s})] \leqslant \hat{O}_N\Big] \geqslant 1-\alpha.$$

And α *is defined as*

$$\alpha = 1 - \int_{\mathbf{x}\in\mathcal{X}'} \frac{1}{\sqrt{(2\pi)^b|\mathbf{\Sigma}^*|}} \exp\left(-\frac{1}{2}\mathbf{x}^T\mathbf{\Sigma}^{*-1}\mathbf{x}\right)d\mathbf{x},$$

where $\mathcal{X}' = \{\mathbf{x} = (\mathbf{P}-\hat{\mathbf{P}})_{\mathcal{S}^+} \Big| D_f(\mathbf{P}||\hat{\mathbf{P}}) \leqslant \tau\}$*. We use* $(\mathbf{P}-\hat{\mathbf{P}})_{\mathcal{S}^+}$ *to denote that we only keep the dimensions that appear in set* $\mathcal{S}^+$*. Therefore,* $\mathbf{x}$ *has b components/dimensions in total.*

We only keep the dimensions in $\mathcal{S}^+$ because of the absolutely continuous property associated with the f-divergence. Intuitively, it means that if the empirical probability of one scenario is asymptotically zero, then the true probability of this empirical probability is also zero.

2. p **-Wasserstein distance-based ambiguity set.** We define $\|\mathbf{P}-\hat{\mathbf{P}}\|$ as $W_p(\mathbf{P},\hat{\mathbf{P}})$ representing the p-Wasserstein distance defined in Definition 1.

$$\mathcal{P}' = \left\{ \mathbf{P} = \{\mathrm{P}(\mathbf{s}), \forall \mathbf{s}\in\mathcal{S}\} : \begin{array}{l} \sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s}) = 1, \\ \mathrm{P}(\mathbf{s}) \geqslant 0, \quad \forall \mathbf{s}\in\mathcal{S}, \\ W_p(\mathbf{P},\hat{\mathbf{P}}) \leqslant \tau. \end{array} \right\} \tag{12}$$

Definition 1. (**p-Wasserstein metric**) The p-Wasserstein distance ($p\in[1,+\infty)$) between distribution $\mathbf{P}$ and $\hat{\mathbf{P}}$ supported on Ξ is defined as

$$W_p(\mathbf{P},\hat{\mathbf{P}}) := \inf\{\left(\int_{\Xi^2} d^p(\xi,\xi)\Pi(d\xi,d\xi)\right)^{\frac{1}{p}} : \Pi \text{ is a joint distribution of } \xi \text{ and } \xi' \text{ with marginals } \mathbf{P} \text{ and } \hat{\mathbf{P}}\},$$

where d' is a metric on Ξ (for example, L2 norm).

Wasserstein distance is studied widely in DRO literature (Esfahani & Kuhn, 2018; Gao & Kleywegt, 2016) recently. It is shown to have superior performances with tractable reformulation in a number of cases. We conclude its reformulation for our model below.

Proposition 4 (Gao et al. (2017))**.** *With p-Wasserstein distance-based ambiguity set, Model (4) is equivalent to*

$$\min_{\mathbf{x}\in\mathcal{X},\lambda>0} \left\{\lambda\tau^p + \sum_{\mathbf{s}\in\mathcal{S}} \hat{\mathrm{P}}(\mathbf{s})\left[\sup_{\mathbf{s}'\in\mathcal{S}}\{Q(\mathbf{x},\mathbf{s}') - \lambda d'^p(\mathbf{s}',\mathbf{s})\}\right]\right\} \tag{13}$$

As shown above, the tractability of using p-Wasserstein distance-based ambiguity set depends on if we can solve

$$\sup_{\mathbf{s}'\in\mathcal{S}}\{Q(\mathbf{x},\mathbf{s}') - \lambda d'^p(\mathbf{s}',\mathbf{s})\}$$

efficiently. This further depends on the structure of the support $\mathcal{S}$ and the choice of metric d', which is beyond the scope of this study.

3. **Ellipsoid ambiguity set.** We define a family of ambiguity sets called ellipsoid ambiguity sets. We make one regularization assumption here to simplify the notations used. We assume the nominal distribution $\hat{\mathbf{P}}$ contains no zero-value components. We will explain later why this does not limit the generality. In this case, we measure the distance between $\mathbf{P}$ and $\hat{\mathbf{P}}$, i.e., $\|\mathbf{P}-\hat{\mathbf{P}}\|$, through

$$(\mathbf{P}-\hat{\mathbf{P}})^T\mathbf{\Sigma}^{-1}(\mathbf{P}-\hat{\mathbf{P}}).$$

Recall that $\mathbf{\Sigma}^{-1}$ is the inverse of the observed information matrix derived in Section 4.3.

$$\mathcal{P}' = \left\{ \mathbf{P} = \{\mathrm{P}(\mathbf{s}), \forall \mathbf{s}\in\mathcal{S}\} : \begin{array}{l} \sum_{\mathbf{s}\in\mathcal{S}} \mathrm{P}(\mathbf{s}) = 1, \\ \mathrm{P}(\mathbf{s}) \geqslant 0, \quad \forall \mathbf{s}\in\mathcal{S}, \\ (\mathbf{P}-\hat{\mathbf{P}})^T\mathbf{\Sigma}^{-1}(\mathbf{P}-\hat{\mathbf{P}}) \leqslant \tau. \end{array} \right\} \tag{14}$$

We want to point out although many existing studies have also proposed ellipsoid ambiguity sets (Delage & Ye, 2010; Li, Mathieu, & Jiang, 2018; Wiesemann, Kuhn, & Sim, 2014), they mostly define it through the support by assuming the first ($\boldsymbol{\mu}$) and second moment information ($\mathbf{\Sigma}_0$) are known or partially known. The ellipsoid in our problem is different from them because it is defined in the probability space. For more discussion, please refer to Appendix I.

Finally, we also prove that DRO with this kind of ellipsoid ambiguity set is still tractable. The result is summarized in Proposition 5, which is a second-order conic programming.

Proposition 5. *The DRO with ellipsoid ambiguity set is equivalent to*

$$\begin{array}{ll} \min\limits_{\mathbf{x},y,\boldsymbol{\beta},\gamma} & y + \hat{\mathbf{P}}^T\mathbf{Q}(\mathbf{x}) + \hat{\mathbf{P}}^T\boldsymbol{\beta} \\ s.t. & \mathbf{x}\in\mathcal{X}, \\ & \beta_{\mathbf{s}} \geqslant 0, \forall \mathbf{s}\in\mathcal{S} \\ & y \geqslant 0, \\ & y^2 \geqslant \tau\big(\mathbf{Q}(\mathbf{x}) - \boldsymbol{\gamma} + \boldsymbol{\beta}\big)^T\mathbf{\Sigma}\big(\mathbf{Q}(\mathbf{x}) - \boldsymbol{\gamma} + \boldsymbol{\beta}\big), \end{array} \tag{15}$$

where we use $\boldsymbol{\gamma}$ *to represent an* $|\mathcal{S}|$*-dimension vector whose all elements equal* γ*. For simplicity, we use* $\mathbf{Q}(\mathbf{x})$*, which is an* $|\mathcal{S}|$*-dimension vector, to denote the vector form of* $Q(\mathbf{x},\mathbf{s})$*.*

Proof. Please refer to Appendix I for detailed proofs. □

5. Computational study

In this section, we conduct computational studies to validate the superiority of the proposed model in some real-world applications. We first study a multi-item two-stage inventory control problem based on the synthetic data in Section 5.1. Please refer to Appendix H for extensions of our model to two-stage stochastic programming. In this experiment, we show the improvements in the out-of-sample performances of our models compared to a data-imputation-based approach. In Section 5.2, we study a portfolio optimization problem based on the real-world historical returns of exchange-traded funds (ETFs) and the US central bank (FED) rate of return from 2006 to 2016 (Boyd, 2019), where our approaches also achieve better out-of-sample performance consistently.

5.1. Multi-item two-stage inventory control problem

5.1.1. Problem formulation

Multi-item two-stage inventory control problem (Axsäter, Schneeweiss, & Silver, 2012) can be described in two periods. At the beginning of the second period, the decision-maker observes

m product demands $\mathbf{b} \in \mathbb{R}^m$ following an unknown joint distribution $\mathbf{P}$. The demand of product i can be served by either placing an order with unit cost a_{1i} in the first period, which will be delivered at the beginning of the second period, or by placing an order with unit cost a_{2i} ($a_{2i} > a_{1i}$) at the beginning of the second period which will be delivered immediately. The excess product units after the second period incur a unit holding cost h_i. If there is a shortfall in the available quantity, then a unit outstocking cost p_i is incurred. We denote the level of inventory for each item by $I_i(\mathbf{b})$, $i = 1, \ldots, m$. The decision-maker wishes to determine the first-stage and second-stage ordering quantities, x_{1i} and $x_{2i}(\mathbf{b})$, for all products to minimize the total ordering, backlogging and holding costs. The problem can be formulated as the following two-stage optimization problem in (16).

$$\begin{aligned} \min \quad & \sum_{i=1}^{m} a_{1i}x_{1i} + \mathbb{E}\left[\sum_{i=1}^{m} a_{2i}x_{2i}(\mathbf{b}) + \sum_{i=1}^{m} \max\{-p_i I_i(\mathbf{b}), h_i I_i(\mathbf{b})\}\right] \\ s.t. \quad & I_i(\mathbf{b}) = x_{1i} + x_{2i}(\mathbf{b}) - b_i, i = 1, \ldots, m, \\ & x_{1i} \geqslant 0, i = 1, \ldots, m, \\ & x_{2i} \geqslant 0, i = 1, \ldots, m, \end{aligned} \tag{16}$$

Correspondingly, our model for (16) under the incomplete data is formulated as

$$\begin{aligned} \min \quad & \sum_{i=1}^{m} a_{1i}x_{1i} + \max_{\mathbf{P}\in\mathcal{P}'} \mathbb{E}_{\mathbf{P}}\left[\sum_{i=1}^{m} a_{2i}x_{2i}(\mathbf{b}) \right. \\ & \left. + \sum_{i=1}^{m} \max\{-p_i I_i(\mathbf{b}), h_i I_i(\mathbf{b})\}\right] \\ s.t. \quad & I_i(\mathbf{b}) = x_{1i} + x_{2i}(\mathbf{b}) - b_i, i = 1, \ldots, m, \\ & x_{1i} \geqslant 0, i = 1, \ldots, m, \\ & x_{2i} \geqslant 0, i = 1, \ldots, m, \end{aligned} \tag{17}$$

where the ambiguity set $\mathcal{P}'$ is defined through (8). Its tractable reformulation is discussed in Appendix H.

5.1.2. Computational results

We compare our model with the data-imputation-based approach. We consider $m = 3$ products whose demands are defined as $D_i = D_{1i} + D_0, \quad i = 1, 2, 3$, where D_0 follows a Poisson distribution, Pois(10), and D_{1i} follows a Poisson distribution Pois$(5i + 5)$. We first randomly generate 300 samples to serve as the known support in this problem. Then, we conducted 100 experiments by generating different random samples. In each experiment, we randomly generate 10 joint data and assume each component will be missed with a probability 0.5. We set the inventory parameters as follows, $a_{i1} = 2$, $a_{2i} = 300$, $h_i = 1$, and $p_i = 240$. The nominal distribution is obtained based on Proposition 1 through CVXOPT.

To evaluate the quality of our models, we draw the boxplots of the out-of-sample performances of the proposed models under different deviation tolerance, τ. We also compare them with one data-imputation-based approach. Because the support of the random variable is assumed to be known, each incomplete data is imputed to the nearest support with respect to the L1 norm in this approach. Then, the optimal solution is obtained based on the empirical distribution of the complete data set. We conclude the results in Fig. 1, where the vertical axis represents the out-of-sample inventory costs.

Results. With partially observed data, our model has similar performances as the data-imputation-based approach when $\tau = 0$. This is because when $\tau = 0$, our method is equivalent to an imputation method based on MLE. However, when $\tau = 0.1$ or 1, the performances are improved significantly, where the 0.75 percentile is greatly reduced and, 0.25 percentile remains stable. In general, our model achieves much lower average inventory costs.

5.2. Portfolio optimization

We also benchmark the proposed approaches for portfolio optimization, where the missing data is a common problem (Radulescu, 2013; Taylor, 2006). In the subsequent numerical experiments, we show that our approach outperforms the data-imputation-based approach on the real-world data set. The experimental data come from the historical returns of exchange-traded funds (ETFs) and the US central bank (FED) rate of return (Boyd, 2019). The data set roughly covers ten years, from January 2006 to December 2016, and includes 2520 data for each asset.

We define the model of portfolio optimization in Section 5.2.1. The experimental settings are presented in Section 5.2.2, which introduces the data set and the implemented methods. Section 5.2.3 concludes the results and discussions.

5.2.1. Mean-risk portfolio optimization model under missing data

Mean-risk portfolio optimization (Esfahani & Kuhn, 2018) considers m assets with returns captured by a random vector $\boldsymbol{\xi} = [\xi_1, \ldots, \xi_m]$ whose joint distribution is $\mathbf{P}$. A portfolio is denoted by a vector $\mathbf{x} = [x_1, \ldots, x_m]$, where

$$\mathbf{x} \in \mathcal{X} = \left\{\mathbf{x} \in \mathbb{R}^m_+ \middle| \sum_{i=1}^{m} x_i = 1\right\}.$$

Each x_i represents the percentage of the investment in asset i for each $1 \leqslant i \leqslant m$. The objective function aims to minimize a weighted sum of the mean of negative returns and the conditional value-at-risk as shown in (18), where $\rho > 0$:

$$\min_{\mathbf{x}\in\mathcal{X}} \quad \mathbb{E}_{\mathbf{P}}\big[\langle -\boldsymbol{\xi}, \mathbf{x}\rangle\big] + \rho \mathrm{CVaR}_{\eta}\big(\langle -\boldsymbol{\xi}, \mathbf{x}\rangle\big) \tag{18}$$

CVaR at level η, $0 < \eta \leqslant 1$ represents the average of the $\eta \times 100\%$ worst portfolio losses. Replacing CVaR in (18) with its definition (Rockafellar, Uryasev et al., 2000), we have

$$\min_{\mathbf{x}\in\mathcal{X}, d\in\mathbb{R}} \quad \mathbb{E}_{\mathbf{P}}\big(\langle -\boldsymbol{\xi}, \mathbf{x}\rangle\big) + \rho\Big[d + \frac{1}{\eta}\mathbb{E}_{\mathbf{P}}(\langle -\boldsymbol{\xi}, \mathbf{x}\rangle - d)^+\Big]. \tag{19}$$

When only incomplete observations of $\boldsymbol{\xi}$ are available, we aim to solve Optimization (20):

$$\min_{\mathbf{x}\in\mathcal{X}, d\in\mathbb{R}} \quad \max_{\mathbf{P}\in\mathcal{P}'} \quad \mathbb{E}_{\mathbf{P}}(\langle -\boldsymbol{\xi}, \mathbf{x}\rangle) + \rho\Big[d + \frac{1}{\eta}\mathbb{E}_{\mathbf{P}}(\langle -\boldsymbol{\xi}, \mathbf{x}\rangle - d)^+\Big]. \tag{20}$$

5.2.2. Numerical settings

Data preprocessing. In the preprocessing step, we round each data to the nearest thousandth and remove the data of the first 500 days because these data follow very different patterns due to the well-known financial crisis of 2007–2008, as shown in Fig. 2.

Training set and test set. We explain the way to generate the training set and test set. The training set is assumed to be known to the decision-makers, and the test set is assumed to be unknown to the decision-makers and is used to evaluate the out-of-sample performance. We model the support based on previous years' data. That is we approximate the known support $\mathcal{S}$ of the returns $\boldsymbol{\xi}$ with the first 300 data. We iteratively use the data from day $[301 + 30 \times (i - 1)]$ to day $(300 + 30 \times i)$ (approximately one month) as the training set, $1 \leqslant i \leqslant 57$. The rest of the data are used as the test set. Therefore, we obtain 57 pairs of the training set and test set.

We use the first $m = 10$ assets in the original data set for all the experiments. Therefore, the data set used in the experiments contains 10 assets with 2020 daily returns each. To model the partially observed data, we assume each dimension of the data in the training set will be missing with a fixed probability 0.5.

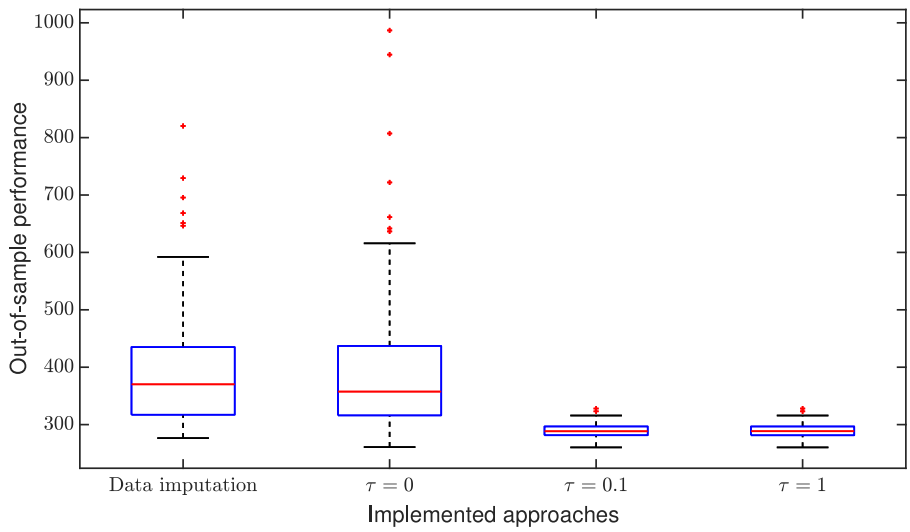


Fig. 1. Out-of-sample performance.

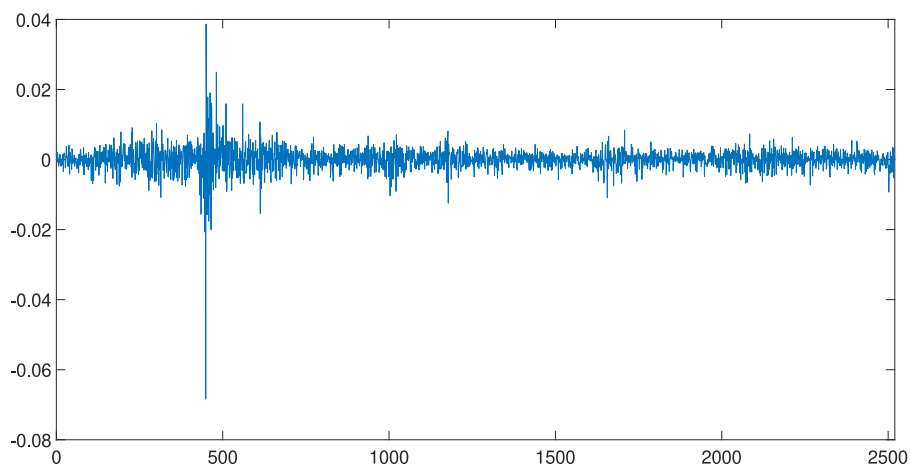


Fig. 2. Daily returns of Asset #1 (iShares Core U.S. Aggregate Bond ETF).

Implemented approaches. We aim to compare Model (20) to a classic data-imputation-based procedure. We explain the implementation details of each approach in the following. The data-imputation-based procedure first recovers the unknown distribution using nearest neighbor imputation; then it solves Model (19) through the empirical distribution based on the completed data set. In the first step because the support $\mathcal{S}$ is known, each incomplete data is imputed to the nearest support with respect to the L1 norm. Besides, if several supports achieve the smallest distance at the same time, one random support among them is used. The nominal distribution used in the proposed model is obtained based on Proposition 1 through CVXOPT.

5.2.3. Results

We conclude the main numerical results on the partially observed data in this section, where our model achieves better out-of-sample performance than the data-imputation-based procedure. We conducted 10 groups of experiments under different values of τ, where $\tau = 0.01, 0.02, \ldots, 0.1$. Each group of experiments contains 57 experiments as discussed in Section 5.2.2. We present the out-of-sample performance in the following.

Out-of-sample performance. The out-of-sample performance is defined as the value of

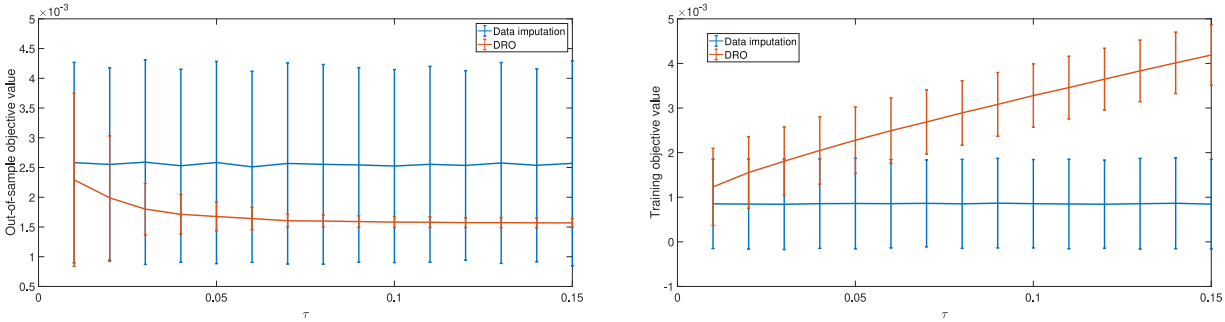
$$\min_{d\in\mathbb{R}} \quad \mathbb{E}_{\hat{\mathbf{P}}}\big(\langle -\boldsymbol{\xi}, \hat{\mathbf{x}}\rangle\big) + \rho\Big[d + \frac{1}{\eta}\mathbb{E}_{\hat{\mathbf{P}}}(\langle -\boldsymbol{\xi}, \hat{\mathbf{x}}\rangle - d)^+\Big], \tag{21}$$

where $\hat{\mathbf{P}}$ represents the empirical distribution of the test set, and $\hat{\mathbf{x}}$ represents the portfolio obtained from corresponding models based on the training set. We repeat each experiments for 10 times (random missing components) to obtain the standard deviation. The results are concluded in Fig. 3a. The optimal training objective values are also presented in Fig. 3b.

Our model outperforms consistently by achieving lower values in (19) and smaller standard deviation. As τ increases, the performance of our model first increases and then stays stable. The standard deviations are also decreasing, which indicates fewer fluctuations in the costs in practice. The results indicate that a good choice of τ should be larger than 0.07 meaning decision makers should consider the worst-case scenario for this problem based on the current data set.

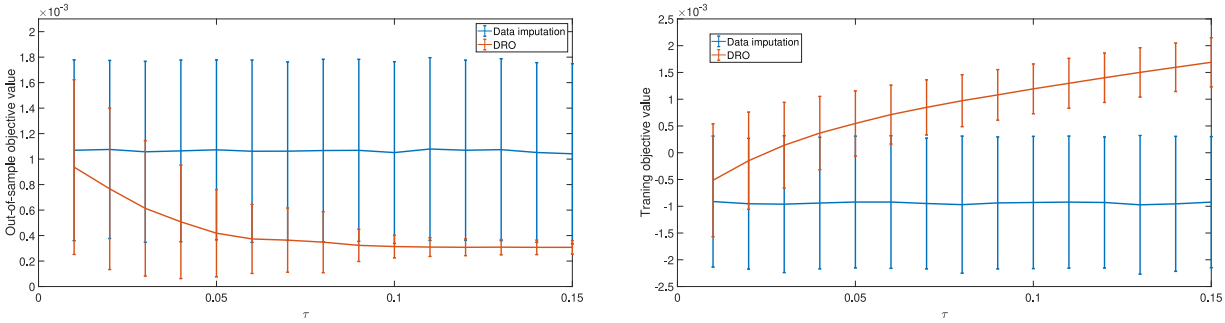
5.2.4. Evaluating the performance for various experiment settings

In this section, we vary the choices of the parameters in the above problem to evaluate the performance of the proposed method. We first change the trade-off parameter ρ to 0.1 (a smaller value). The results are concluded in Fig. 4a and b. From the results, we can see that the difference between the data-imputation-based method and our method becomes smaller. This is because the robustness is less appreciated when ρ becomes smaller.



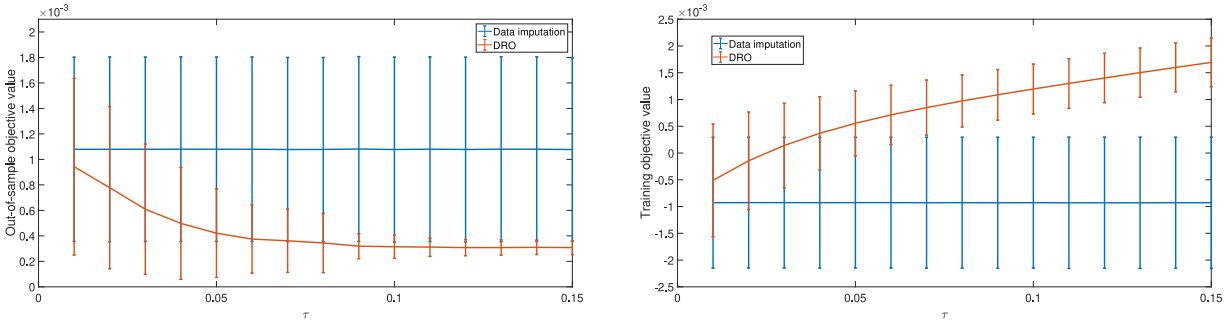
(a) Out-of-sample objective value (b) Training objective value

Fig. 3. Out-of-sample objective values and training objective values under different values of τ.



(a) Out-of-sample objective value (b) Training objective value

Fig. 4. Smaller trade-off parameter ρ.



(a) Out-of-sample objective value (b) Training objective value

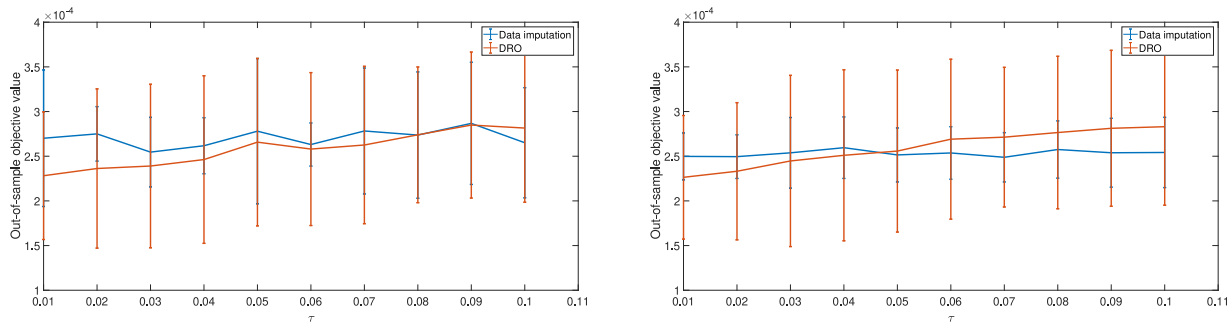
Fig. 5. Small missing probability.

Next, we keep the parameter $\rho = 0.1$ and set the missing probability to 0.1. The results are summarized in Fig. 5a and b. Both the data-imputation method and our method perform a little worse than that when the missing probability is 0.5. However, this difference is very small. To better illustrate the effects of the missing probability, we increase the number of observed data to decrease the uncertainties brought by the finite sample size in Fig. 6a and b. We increase the sample size of the training set to 800. We iteratively use the data from day $[301 + 800 \times (i1)]$ to day $(300 + 800 \times i)$ as the training set, $1 \leqslant i \leqslant 2$. The rest of the data are used as the test set. It can be concluded that when the missing probability is small, the proposed model has less advantage compared to data imputation-based methods (resulting in higher objective values than that of imputation-based methods with the same choice of τ compared to models with high missing probability).

5.2.5. Illustrations of ellipsoid ambiguity set

Finally, we also showcase the L1-norm-based model and model with the ellipsoid ambiguity set. Because of the tractability issue, we only considered a case where the parameter ρ is zero. The other parameters follow the settings described in the last section. As shown in Fig. 7, the ellipsoid-based method is more robust than the L1-norm-based method when τ is small. Additionally, when τ becomes large, it is less conservative compared to the L1-norm-based method.

In general, it is challenging (even for problems with complete data) to claim that one ambiguity set is better than the others. Because the performance is very problem and dataset dependent, we recommend practitioners to choose their ambiguity sets according to the settings of their problem. For example, f-divergence is suitable for discrete cases because of the absolutely continuous prop-



(a) High missing probability (=0.5). (b) Small missing probability (=0.1).

Fig. 6. Large sample size (800).

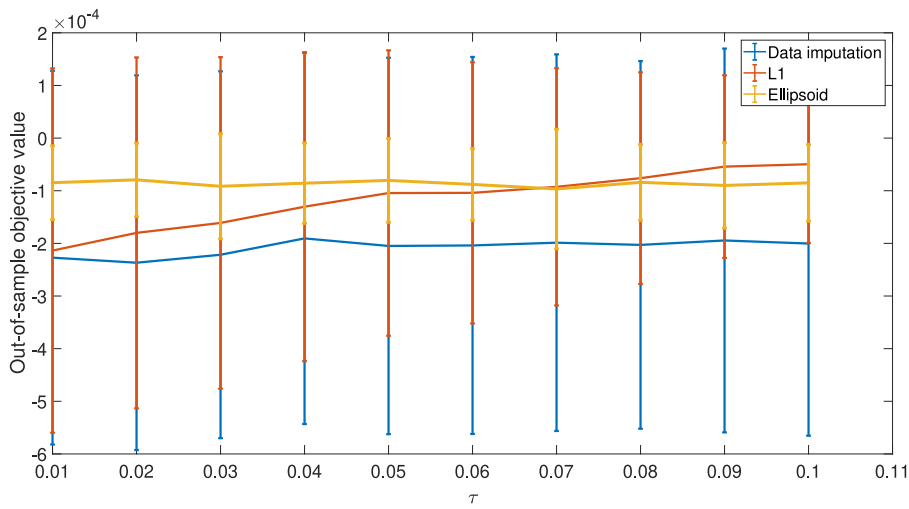


Fig. 7. Comparisons with Ellipsoid ambiguity set.

erty. For nominal data, the L1-norm-based method is optimal from the modeling perspective (see Appendix A).

6. Conclusion

This paper develops a new DRO framework for data-driven stochastic optimization facing incomplete data sets. Our model represents an integrated analysis of missing data and stochastic optimization, which is different from the most popular data-imputation-based approaches. We provide theoretical guarantees and utilize the concepts in MLE and the observed information matrix to obtain the bounds for the performance. Several classes of ambiguity sets with their reformulations are discussed. There are several directions for future work. First, constructing ambiguity sets to address missing data for random variables with continuous support is of great interest to study. Specifically, a certain framework can be developed for parametric models, where the unknown distribution is assumed to come from some continuous distribution family. If we can also quantify the uncertainties coming from the missing data and incorporate them into a robust/DRO framework, then the corresponding model is applicable in continuous cases. Second, developing a new optimization algorithm with appropriate scenario reduction methods can improve the proposed model's efficiency and tractability when the number of scenarios is very large.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2022.06.032.

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